

Jeyaseelan Krishna Udiyappan

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EDUCATION

Ph.D. in Chemical Engineering

Ohio University, Ohio, USA

GPA: 3.84/4.00

2024 – Present

- Research focus: mathematical modeling and design of electrochemical slurry reactors, complex-fluid transport, and multiphysics simulation.

B.Tech. in Chemical and Electrochemical Engineering

CSIR–Central Electrochemical Research Institute (CECRI), Tamil Nadu, India

CGPA: 8.76/10

2019 – 2023

RESEARCH EXPERIENCE

Ohio University

Advisor: Dr. Madhu V. Majji

2024 – Ongoing

– Electrochemical Slurry Reactors and Complex-Fluid Transport

- Investigate coupled rheology, mass/ionic transport, electronic transport, and electrochemical kinetics in flowable suspension electrodes to establish physics-based reactor design principles.
- Develop Couette, rotating-disk, and flow-cell experiments to quantify shear-dependent mass transfer in conductive and non-conductive suspensions using electrochemical limiting-current methods.
- Characterize carbon-black and fumed-silica suspensions using rheometry and Herschel–Bulkley constitutive models to relate yield stress and shear thinning to transport behavior.
- Develop continuum electrochemical models coupling porous-electrode theory, non-Newtonian flow, shear-dependent mass transfer, and electronic conductivity to predict polarization, current distribution, pumping losses, and reactor performance.

– Fundamental Yield-Stress Rotating Flows

- Develop Python/MATLAB nonlinear boundary-value solvers and COMSOL finite-element models for Newtonian, Bingham, and Herschel–Bulkley rotating flows.
- Analyze confinement, similarity breakdown, multiple solution branches, yielded/unyielded regions, and local hydrodynamic scaling relevant to rotating-electrode transport.

– Experimental Systems Development

- Designed and fabricated custom electrochemical flow cells using SolidWorks, resin additive manufacturing, Nomad CNC machining, and graphite milling, progressing from CAD to assembly and electrochemical testing.
- Operate and maintain rheometry/electrochemical instrumentation and serve as lab in-charge for rheometer operations.

CSIR–Central Electrochemical Research Institute (CECRI)

Advisor: Dr. M. Sathish

2021 – 2023

– Zinc-Ion Battery Research

- Recovered ZnMn_2O_4 from end-of-life alkaline-battery materials and prepared carbon composites for aqueous zinc-ion battery electrodes.
- Evaluated cycling stability and electrochemical performance using galvanostatic charge/discharge and complementary electrochemical characterization.

– Advanced Materials for Batteries and Supercapacitors

- Synthesized manganese-based cathode materials for lithium- and sodium-ion batteries and fabricated carbon-based symmetric cylindrical supercapacitors.
 - Characterized materials using XRD and FE-SEM and evaluated electrochemical response using cyclic voltammetry and galvanostatic cycling.
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SELECTED COMPUTATIONAL PROJECTS

Coupled Fluid–Particle Heat Transfer in a Serpentine Microtube – ANSYS Fluent/DPM

- Built a coupled CFD–Discrete Phase Method model with two-way momentum and energy exchange; evaluated mesh quality, laminar/turbulent regimes, particle trajectories, concentration fields, and heat-transfer response.

Corrosion Under Insulation – COMSOL Multiphysics

- Modeled oxygen transport and corrosion behavior in thin electrolyte films using finite-element multiphysics methods.
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INTERNSHIP

Institute of Chemical Technology (ICT), Mumbai, India

July – August 2022

Advisor: Dr. Paresh H. Salame

Topic: Synthesis of NASICON-based polyanionic cathode material for sodium-ion batteries through an ultrasound-assisted wet-chemical (sol–gel) route.

SKILLS AND EXPERTISE

TRANSPORT AND MODELING: Transport phenomena, electrochemical kinetics, porous-electrode theory, mass transfer, non-Newtonian/yield-stress flow, Herschel–Bulkley/Bingham rheology, nonlinear ODE/BVP methods, finite-element modeling, CFD, particle transport.

EXPERIMENTAL: Rheometry, electrochemical flow cells, rotating-disk electrochemistry, chronoamperometry, cyclic voltammetry, galvanostatic cycling, suspension preparation, XRD, FE-SEM.

SCIENTIFIC COMPUTING: Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, COMSOL Multiphysics, ANSYS Fluent.

ENGINEERING AND PROTOTYPING: SolidWorks, resin 3D printing, Nomad CNC machining, graphite milling, custom cell/reactor fabrication, experimental setup commissioning.

CONFERENCE CONTRIBUTIONS AND PRESENTATIONS

- **248th Electrochemical Society Meeting (2025):** Oral presentation, “Ionic Mass Transport in Slurry Electrodes.” Presented by J. K. Udiyappan.
 - **J. K. Udiyappan, M. V. Majji,** “Performance Tradeoffs in Redox Flow Batteries with Slurry Electrodes,” *Electrochemical Society Meeting Abstracts* 248, 2754 (2025). Presented by M. V. Majji.
 - **J. K. Udiyappan, M. V. Majji,** “Charge Transport Trade-offs in Colloidal Suspension Electrodes under Flow,” 96th Annual Meeting of the Society of Rheology, Santa Fe, NM (2025). Presented by M. V. Majji.
 - **3-Minute Thesis (3MT):** “Designing Slurry Electrodes for Large-Scale Electrochemical Reactors.”
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LEADERSHIP AND OUTREACH

- As the first PhD student in a newly formed research group, help establish experimental infrastructure through equipment procurement, safety coordination, instrument commissioning and maintenance, vendor support, and laboratory workflow development.
- Mentor two undergraduate researchers in electrochemical analysis, suspension rheology, experimental methods, and data interpretation.
- Volunteer for middle-school K–12 science-fair activities, supporting student scientific communication and evaluation.